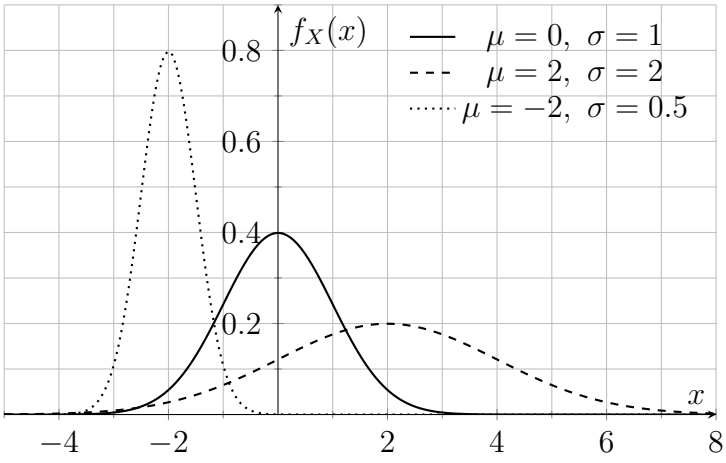
Last Time

- **Gaussian (normal) Random Variable** $X \sim N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2},$$



- **Standard Gaussian (normal) Random Variable** $X \sim N(0, 1)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

If Z is a standard normal random variable, we denote its CDF, called **standard normal CDF**, as

$$\Phi(z) = F_Z(z) = \int_{-\infty}^{z} f_Z(x)dx = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

It has the following property:

$$\Phi(-z) = 1 - \Phi(z).$$

If $X \sim N(\mu, \sigma^2)$, we have

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

$\Phi(-z) = 1 - \Phi(z)$

If $X \sim N(\mu, \sigma^2)$, we have

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

The standard normal CDF $\Phi(z)$

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.97725	2.50	0.99379
0.01	0.5040	0.51	0.6950	1.01	0.8438	1.51	0.9345	2.01	0.97778	2.51	0.99396
0.02	0.5080	0.52	0.6985	1.02	0.8461	1.52	0.9357	2.02	0.97831	2.52	0.99413
0.03	0.5120	0.53	0.7019	1.03	0.8485	1.53	0.9370	2.03	0.97882	2.53	0.99430
0.04	0.5160	0.54	0.7054	1.04	0.8508	1.54	0.9382	2.04	0.97932	2.54	0.99446
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.05	0.97982	2.55	0.99461
0.06	0.5239	0.56	0.7123	1.06	0.8554	1.56	0.9406	2.06	0.98030	2.56	0.99477
0.07	0.5279	0.57	0.7157	1.07	0.8577	1.57	0.9418	2.07	0.98077	2.57	0.99492
0.08	0.5319	0.58	0.7190	1.08	0.8599	1.58	0.9429	2.08	0.98124	2.58	0.99506
0.09	0.5359	0.59	0.7224	1.09	0.8621	1.59	0.9441	2.09	0.98169	2.59	0.99520
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.10	0.98214	2.60	0.99534
0.11	0.5438	0.61	0.7291	1.11	0.8665	1.61	0.9463	2.11	0.98257	2.61	0.99547
0.12	0.5478	0.62	0.7324	1.12	0.8686	1.62	0.9474	2.12	0.98300	2.62	0.99560
0.13	0.5517	0.63	0.7357	1.13	0.8708	1.63	0.9484	2.13	0.98341	2.63	0.99573
0.14	0.5557	0.64	0.7389	1.14	0.8729	1.64	0.9495	2.14	0.98382	2.64	0.99585
0.15	0.5596	0.65	0.7422	1.15	0.8749	1.65	0.9505	2.15	0.98422	2.65	0.99598
0.16	0.5636	0.66	0.7454	1.16	0.8770	1.66	0.9515	2.16	0.98461	2.66	0.99609
0.17	0.5675	0.67	0.7486	1.17	0.8790	1.67	0.9525	2.17	0.98500	2.67	0.99621
0.18	0.5714	0.68	0.7517	1.18	0.8810	1.68	0.9535	2.18	0.98537	2.68	0.99632
0.19	0.5753	0.69	0.7549	1.19	0.8830	1.69	0.9545	2.19	0.98574	2.69	0.99643
0.20	0.5793	0.70	0.7580	1.20	0.8849	1.70	0.9554	2.20	0.98610	2.70	0.99653
0.21	0.5832	0.71	0.7611	1.21	0.8869	1.71	0.9564	2.21	0.98645	2.71	0.99664
0.22	0.5871	0.72	0.7642	1.22	0.8888	1.72	0.9573	2.22	0.98679	2.72	0.99674
0.23	0.5910	0.73	0.7673	1.23	0.8907	1.73	0.9582	2.23	0.98713	2.73	0.99683
0.24	0.5948	0.74	0.7704	1.24	0.8925	1.74	0.9591	2.24	0.98745	2.74	0.99693
0.25	0.5987	0.75	0.7734	1.25	0.8944	1.75	0.9599	2.25	0.98778	2.75	0.99702
0.26	0.6026	0.76	0.7764	1.26	0.8962	1.76	0.9608	2.26	0.98809	2.76	0.99711
0.27	0.6064	0.77	0.7794	1.27	0.8980	1.77	0.9616	2.27	0.98840	2.77	0.99720
0.28	0.6103	0.78	0.7823	1.28	0.8997	1.78	0.9625	2.28	0.98870	2.78	0.99728
0.29	0.6141	0.79	0.7852	1.29	0.9015	1.79	0.9633	2.29	0.98899	2.79	0.99736
0.30	0.6179	0.80	0.7881	1.30	0.9032	1.80	0.9641	2.30	0.98928	2.80	0.99744
0.31	0.6217	0.81	0.7910	1.31	0.9049	1.81	0.9649	2.31	0.98956	2.81	0.99752
0.32	0.6255	0.82	0.7939	1.32	0.9066	1.82	0.9656	2.32	0.98983	2.82	0.99760
0.33	0.6293	0.83	0.7967	1.33	0.9082	1.83	0.9664	2.33	0.99010	2.83	0.99767
0.34	0.6331	0.84	0.7995	1.34	0.9099	1.84	0.9671	2.34	0.99036	2.84	0.99774
0.35	0.6368	0.85	0.8023	1.35	0.9115	1.85	0.9678	2.35	0.99061	2.85	0.99781
0.36	0.6406	0.86	0.8051	1.36	0.9131	1.86	0.9686	2.36	0.99086	2.86	0.99788
0.37	0.6443	0.87	0.8078	1.37	0.9147	1.87	0.9693	2.37	0.99111	2.87	0.99795
0.38	0.6480	0.88	0.8106	1.38	0.9162	1.88	0.9699	2.38	0.99134	2.88	0.99801
0.39	0.6517	0.89	0.8133	1.39	0.9177	1.89	0.9706	2.39	0.99158	2.89	0.99807
0.40	0.6554	0.90	0.8159	1.40	0.9192	1.90	0.9713	2.40	0.99180	2.90	0.99813
0.41	0.6591	0.91	0.8186	1.41	0.9207	1.91	0.9719	2.41	0.99202	2.91	0.99819
0.42	0.6628	0.92	0.8212	1.42	0.9222	1.92	0.9726	2.42	0.99224	2.92	0.99825
0.43	0.6664	0.93	0.8238	1.43	0.9236	1.93	0.9732	2.43	0.99245	2.93	0.99831
0.44	0.6700	0.94	0.8264	1.44	0.9251	1.94	0.9738	2.44	0.99266	2.94	0.99836
0.45	0.6736	0.95	0.8289	1.45	0.9265	1.95	0.9744	2.45	0.99286	2.95	0.99841
0.46	0.6772	0.96	0.8315	1.46	0.9279	1.96	0.9750	2.46	0.99305	2.96	0.99846
0.47	0.6808	0.97	0.8340	1.47	0.9292	1.97	0.9756	2.47	0.99324	2.97	0.99851
0.48	0.6844	0.98	0.8365	1.48	0.9306	1.98	0.9761	2.48	0.99343	2.98	0.99856
0.49	0.6879	0.99	0.8389	1.49	0.9319	1.99	0.9767	2.49	0.99361	2.99	0.99861

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

Example [Sensor Measurement Error] A temperature sensor has measurement error X (in °C) modeled as a Gaussian random variable with

$$\mu = 0, \qquad \sigma = 0.5.$$

Thus, its PDF is

$$f_X(x) = \frac{1}{0.5\sqrt{2\pi}} \exp\left(-\frac{x^2}{2(0.5)^2}\right).$$

The sensor is considered sufficiently accurate if the magnitude of its measurement error is less than 0.8°C. Find the probability that a measurement is sufficiently accurate.

$|X| < 0.8$

The required probability is

$$P(|X| < 0.8) = P(-0.8 < X < 0.8)$$

$\Phi\left(\frac{x-\mu}{\sigma}\right) = F_X(x)$

$$= F_X(0.8) - F_X(-0.8)$$

$\mu = 0, \quad \sigma = 0.5$

$$= \Phi\left(\frac{0.8-\mu}{\sigma}\right) - \Phi\left(\frac{-0.8-\mu}{\sigma}\right)$$

$$= \Phi(1.6) - \Phi(-1.6)$$

$$= \Phi(1.6) - (1 - \Phi(1.6))$$

$$= 2\Phi(1.6) - 1$$

from the Table $\rightarrow \approx 2 \cdot 0.9452 - 1$

$$= 0.8904$$

tells us how probability is distributed among the possible values of X.

Mixed Random Variables

Mixed Random Variable

A random variable X is of **mixed type** if its distribution contains both a discrete component and a continuous component.

In other words, there is at least one value x such that $P(X = x) > 0$ and the sum of the probabilities of all values x with positive probability is not one.

Example [Received Signal Level] Suppose X represents the received signal level at a wireless receiver. Due to complete signal loss,

$$P(X = 0) = 0.2.$$

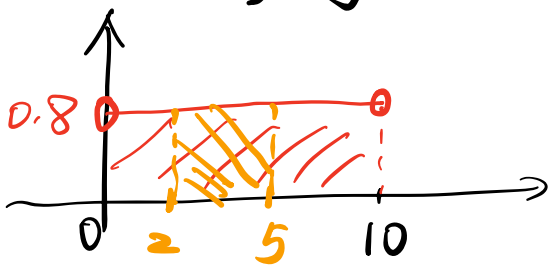
If the signal is successfully received, which occurs with probability 0.8, suppose its level is uniformly distributed over $(0, 10)$. Thus, X has both a discrete component and a continuous component.

(a) Find the probability that the received signal level is between 2 and 5. $2 < X < 5$

(b) Sketch the CDF of X.

(a) $P(2 < X < 5)$

For $0 < X < 10$, the continuous density function is $f_X(x) = 0.8 \frac{1}{10}$



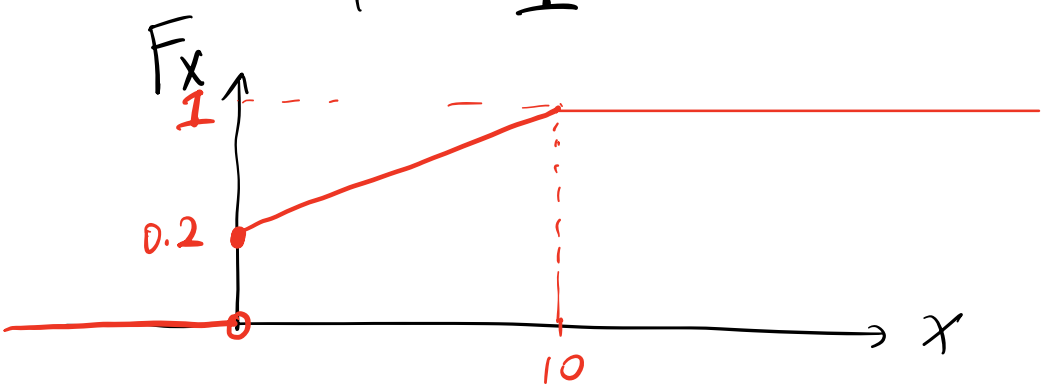
$$P(2 < X < 5) = \int_2^5 f_X(x)\,dx$$

$$= 0.08 \cdot (5 - 2) = 0.24$$

(b)

The CDF of X:

$$F_X(x) = \begin{cases} 0 & , \ x < 0 \\ 0.2 & , \ x = 0 \\ \underbrace{P(X=0) + \int_0^x f_X(x)\,dx}_{0.2 + 0.08x} & , \ 0 < x < 10 \\ 1 & , \ x \geq 10 \end{cases}$$



EXPECTED VALUES

Motivation: repeated experiments

Suppose a **discrete** random variable X can take the values $x_1, x_2, \ldots, x_m$ with probabilities

$$P(X = x_k) = p_k.$$

Imagine repeating the experiment N times. Suppose x_k occurs a_k times, where

$$a_1 + a_2 + \cdots + a_m = N.$$

The average of the N observed values is

$$\begin{aligned} \text{Average} &= \frac{a_1 x_1 + a_2 x_2 + \cdots + a_m x_m}{N} \\ &= x_1 \frac{a_1}{N} + x_2 \frac{a_2}{N} + \cdots + x_m \frac{a_m}{N}. \end{aligned}$$

Here,

$$\frac{a_k}{N}$$

is the relative frequency with which $X = x_k$ occurs. When N is large, we expect

$$\frac{a_k}{N} \approx P(X = x_k) = p_k.$$

Therefore, the observed average should approach

$$x_1 p_1 + x_2 p_2 + \cdots + x_m p_m.$$

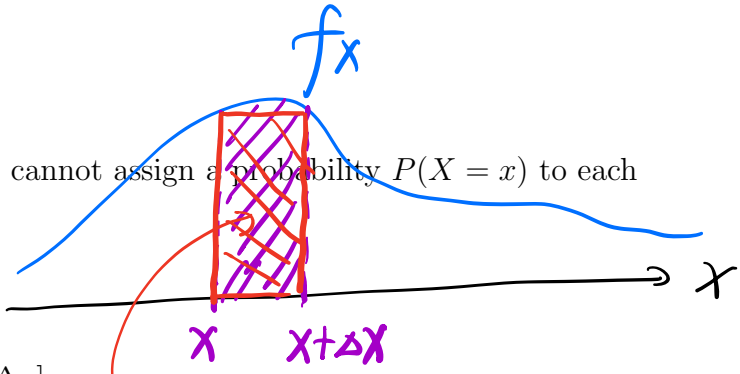
This motivates the expected value:

$$E[X] = \sum_{k=1}^{m} x_k p_k.$$

The expected value of X represents the long-run average value of X when the random experiment is repeated many times.

What if X is continuous?

Individual values now have probability zero, so we cannot assign a probability $P(X = x)$ to each value and add $xP(X = x)$.



Instead, consider a very small interval

$$[x, x + \Delta x].$$

For a continuous random variable with PDF $f_X(x)$,

$$P(x \le X \le x + \Delta x) \approx f_X(x)\Delta x.$$

Values in this small interval are approximately x, so its contribution to the long-run average is approximately

$$x\, f_X(x)\Delta x.$$

Adding the contributions from all small intervals gives

Riemann Sum

$$\sum x\, f_X(x)\Delta x,$$

and as the intervals become arbitrarily small,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

Expected Value

Let X be a random variable. The **expected value (or mean) of X**, denoted by $E[X]$ or μ_X, is defined (*provided it exists*) as

R_X is the range of X.

$$E[X] = \mu_X = \begin{cases} \displaystyle\sum_{x \in \mathcal{R}_X} x\, p_X(x), & X \text{ discrete}, \\ \displaystyle\int_{-\infty}^{\infty} x\, f_X(x)\, dx, & X \text{ continuous}. \end{cases}$$

Interpretation: $E[X]$ is a probability-weighted average of the possible values of X, interpreted as a long-run average over repeated experiments.

Example 1 - Discrete random variable

A fair six-sided die is rolled. Let X be the number obtained. Find $E[X]$.

We know that

$$p_X(x) = P(X = x) = \frac{1}{6} \qquad x = 1, 2, \cdots, 6$$

$$E[X] = \sum_{x=1}^{6} x\, p_X(x)$$

$$= \frac{1}{6}(1 + 2 + 3 + \cdots + 6) = 3.5$$

Note: $E[X]$ does not have to be a possible value of X.

Example 2 - Continuous random variable

Let X be uniformly distributed on $[0, 2]$, with PDF

$$f_X(x) = \begin{cases} \frac{1}{2}, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find $E[X]$.

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

$$= \int_{-\infty}^{0} x f_X(x)\, dx + \int_{0}^{2} x f_X(x)\, dx + \int_{2}^{\infty} x f_X(x)\, dx$$

($f_X(x) = 0$ on the first and third integrals; $f_X(x) = \frac{1}{2}$ on the second)

$$= \int_0^2 \frac{1}{2} x\, dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^2$$

$$= 1$$

Over many independent observations of X, their average approaches 1.